(AP) & gap junctions (19.2)(29) 1φ. 26. 2φ26

cardiac muscle → undergoes aerobic respr. mobilizing lipids & carbs, Myoglobin, lipids, & glycogen are all stored within cytoplasm

Cardiac muscle cells undergo twitch-type contractions with long refractory periods, followed by brief relaxation periods → to prevent tetanyl
↳ tetanyl

Repaired cells don't function well → dead cells are replaced with scar tissue
Mycardial cells produced within the patient (in situ) are nonfunctional

Conduction System of the Heart

↳ are seperated into a petri dish → alive
embryonic heart cells → each is capable of generating it own electrical impulse followed by contraction

How do two independenty beating embryonic cardiac muscle cells are placed?
↳ cells with the higher inherent rate sets the pace → & the impulse spreads from the faster to the slower cell to trigger conc contraction.

-full adult heart maintains the capability of generating its own electrical impulse
↳ triggered by the fastest cell of the cardiac conduction system